

CAT Notes

Sanchay Joshi

2 March 2024

Contents

I	Number System	7
II	Arithmetic	9
1	Alligation and Mixture	11
1.1	See-Saw method of finding weighted average	11

Introduction

This is a series of notes that I am making to learn $\text{\LaTeX}$ as well as makes notes for CAT and any aptitude exam preparation. The youtube channels and resources I am using are as follows

- [Rodha Youtube Channel](#)
- [Free CAT Question Bank](#) : This contains question for each topic.

Part I

Number System

Part II

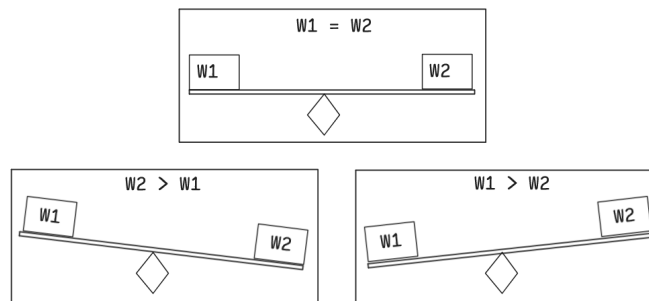
Arithmetic

Chapter 1

Alligation and Mixture

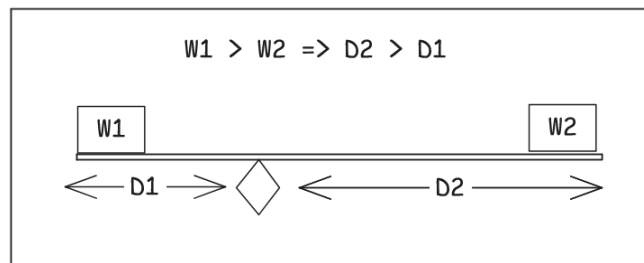
1.1 See-Saw method of finding weighted average

As the name suggests, this is a method that resembles a see-saw. A see-saw is a horizontal bar where at the end, weights are put on. To keep the see-saw balanced, either the weights must be equal or the "torque" must be equal

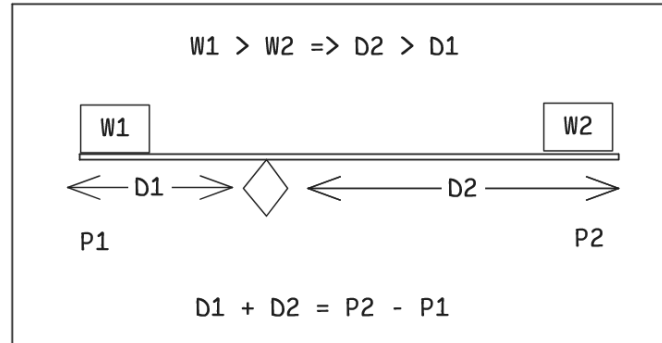


The torque is defined as "Weight * distance from center". The weight w and the distance d from center are inversely proportional to each other

- If $W_1 > W_2$ then $W_1 * d_1 > W_2 * d_2$.
- However, to keep the see-saw in equilibrium, the torque must be same
- To keep the torque same, $d_1 < d_2 \implies W_1 * d_1 = W_2 * d_2$
- We can say that $W \propto \frac{1}{d} \implies W_1 * d_1 = W_2 * d_2 \implies \frac{W_1}{W_2} = \frac{d_2}{d_1}$



Using the ratio of weights, we can determine ratio of distances. To solve our questions, we will put another attribute on weight called "Price". Sum of distances will be equal to the difference of the prices $\Rightarrow d_1 + d_2 = p_2 - p_1$



Question 1.1 : 68Kg of apples are sold at Rs 63/Kg and 85Kg of apples are sold at Rs 81/Kg. Find average price

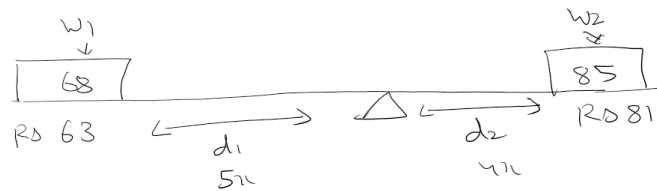
Standard Method

We can find weighted average as follows

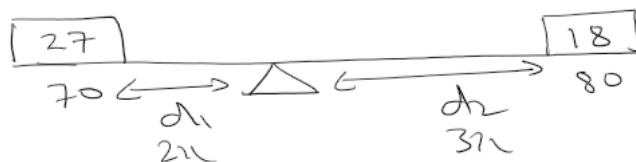
$$\begin{aligned}
 &= \frac{(68 * 63) + (85 * 81)}{68 + 85} \\
 &= \frac{17 * (4 * 63 + 5 * 81)}{17 * (4 + 5)} \\
 &= \frac{252 + 405}{9} \\
 &= 73
 \end{aligned}$$

See-Saw Method

- Weights = 68 Kg, 85 Kg
- Price = Rs 63/Kg, Rs 81/Kg
- $\frac{d_2}{d_1} = \frac{W_1}{W_2} \Rightarrow \frac{68}{85} \Rightarrow \frac{4}{5}$
- Therefore, $d_1 = 5x, d_2 = 4x$
- Now, $d_1 + d_2 = P_2 - P_1 \Rightarrow 9x = 81 - 63 \Rightarrow x = \frac{18}{2} = 2$
- Average = $p_1 + d_1 \Rightarrow 63 + (5 * 2) = 73$



Question 1.2 : 27 boys have 70 marks as average and 18 girls have 80 marks average. Find average of class



- | | | |
|--|--|--|
| <ul style="list-style-type: none"> • Weight = quantities, Price = marks • $W_1 = 27, P_1 = 70$ • $W_2 = 18, P_2 = 80$ | | <ul style="list-style-type: none"> • $\frac{d_2}{d_1} = \frac{W_1}{W_2} \implies \frac{27}{18} \implies \frac{3}{2}$ • $d_1 = 2x, d_2 = 3x$ • $d_1 + d_2 = P_2 - P_1 \implies 5x = 10 \implies x = 2$ • Average = $P_1 + d_1 \implies 70 + (2 * 2) = 74$ |
|--|--|--|